Yinan Huang

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Education

Georgia Institute of Technology, USA, Ph.D. in Machine Learning — Sept 2023 – 2027 (Expected)

- Advisor: Pan Li

Duke University, USA, M.S. in Electrical and Computer Engineering — Sept 2020 - May 2023

- GPA: 4.0/4.0

Sun Yat-sen University, China, B.S. in Physics — Sept 2016 - May 2020

- GPA: 4.3/5.0, Rank: 1/83

Research Interests

Geometric Deep Learning, Generative Models, AI for Science

Research Experience

NEC Laboratories America, Research Intern (Incoming) — Jun 2026 - Aug 2026

Peking University, Research Intern — Feb 2022 – Sept 2022

Beijing Institute for General Artificial Intelligence, Research Intern — Sept 2021 – Feb 2022

Selected Research

Expressive and Generalizable Spectral Graph Learning

- Developed stable and expressive graph positional encodings, addressing the key tension between structural expressivity and model stability, leading to improved robustness and generalization (*ICLR 2024*).
- Proposed multi-potential Magnetic Laplacian positional encodings for directed graphs, addressing the lack of principled methods for modeling directionality and improving performance on direction-sensitive tasks such as circuit analysis (*ICLR 2025*).
- Established a novel combinatorial connection between magnetic graph matrix spectra with random walks on directed graphs, enabling efficient spectral methods for motif detection and link prediction (*PNAS 2026*).

Flow/Diffusion-based Model Sequential Inference Acceleration

- Developed a Bayesian filtering framework for accelerating flow matching, reducing inference steps while maintaining competitive forecasting and decision-making performance (*ICLR Workshop 2026, **Spotlight***).

Geometry-aware Molecular Design

- Built an E(3)-equivariant generative model that jointly learns molecular graphs and 3D geometry, improving structure-aware linker generation for drug discovery (*ICML 2022, **Oral***).

Publications

[1] Accelerated Sequential Flow Matching: A Bayesian Filtering Perspective
Yinan Huang, Hans Hao-Hsun Hsu, Junran Wang, Bo Dai, Pan Li
ICLR Workshop ReALM–GEN, 2026 (Spotlight)

[2] What Can We Learn from State Space Models for Machine Learning on Graphs?
Yinan Huang*, Siqi Miao*, Pan Li
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026

[3] Powers of Magnetic Graph Matrix: Fourier Spectrum, Walk Compression, and Applications
Yinan Huang, David F Gleich, Pan Li
The Proceedings of the National Academy of Sciences (PNAS), 2026

[4] GenAI Copyright Evidence with Operational Meaning

Eli Chien, Amit Saha, ***Yinan Huang***, Pan Li
ICML Workshop on Reliable and Responsible Foundation Models, 2025

[5] Differentially Private Relational Learning with Entity-level Privacy Guarantees
Yinan Huang*, Haoteng Yin*, Eli Chien, Rongzhe Wei, Pan Li
Advances in Neural Information Processing Systems (NeurIPS), 2025.

[6] What Are Good Positional Encodings for Directed Graphs?
Yinan Huang, Haoyu Wang, Pan Li
International Conference on Learning Representations (ICLR), 2025.

[7] On the Stability of Expressive Positional Encodings for Graphs
Yinan Huang*, William Lu*, Joshua Robinson, Yu Yang, Muhan Zhang, Stefanie Jegelka, Pan Li
International Conference on Learning Representations (ICLR), 2024.

[8] Is Distance Matrix Enough for Geometric Deep Learning?
Zian Li, Xiyuan Wang, ***Yinan Huang***, Muhan Zhang
Advances in Neural Information Processing Systems (NeurIPS), 2023.

[9] Boosting the Cycle Counting Power of Graph Neural Networks with I^2-GNNs
Yinan Huang, Xingang Peng, Jianzhu Ma, Muhan Zhang
International Conference on Learning Representations (ICLR), 2023.

[10] 3DLinker: An E(3) Equivariant Variational Autoencoder for Molecular Linker Design
Yinan Huang, Xingang Peng, Jianzhu Ma, Muhan Zhang
International Conference on Machine Learning (ICML), 2022 (Oral).

Honors and Awards

- Travel Award for ICLR 2025
- Georgia Tech ECE Fellowship 2023
- China National Scholarship 2017

Professional Service

- Reviewer for International Conference on Machine Learning (ICML) 2023-2026
- Reviewer for International Conference on Learning Representations (ICLR) 2024-2026
- Reviewer for Advances in Neural Information Processing Systems (NeurIPS) 2023-2026
- Program Committee for Association for the Advancement of Artificial Intelligence (AAAI) 2026
- Reviewer for Association for Computing Machinery's Special Interest Group on Knowledge Discovery and Data Mining (KDD) 2026
- Teaching Assistant: ECE 3077 Introduction to Probability and Statistics, ECE 6250 Advanced Digital Signal Processing

Skills

- Programming languages and frameworks: Python, Pytorch, Pytorch Geometric, Matlab, C